INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

TAPE, LACING AND TYING, TFE-FLUOROCARBON

The General Services Administration has authorized the use of this commercial item description as a replacement for Type III of MIL-T-43435B for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers TFE-fluorocarbon (tetra fluorocarbon) tape, flat braided, for lacing and tying.

2. CLASSIFICATION. Tape shall conform to the sizes and finishes as specified. (see 3.3 and 3.6)

2.1 Size. The size shall be identified as 2, 4 and 5. (see Table 1)

2.2 Finish. The finish shall be identified as A or C. (see 3.6)

3. SALIENT CHARACTERISTICS.

3.1 Material. The yarn used in the fabrication of the tape shall be a high tenacity, continuous filament, TFE-fluorocarbon yarn.

3.2 Construction. The tape shall be a flat braid construction.

3.3 Physical requirements. Tape shall conform to the requirements of Table 1.

TABLE 1: Tape, Lacing and Tying, TFE-Fluorocarbon

Size	Width in inches		Thickness in inches		Breaking strength
	Min	Max	Min	Max	(Lbs, min)
2	.108	.132	.009	.014	30
4	.059	.072	.009	.014	15
5	.023	.028	.009	.014	10

3.4 Elongation. Maximum elongation at breaking strength shall not be greater than 30% when tested as specified in Fed-STD-191 Method 4108. This test to be performed simultaneously with breaking strength.

3.5 Color. Unless otherwise specified (see 7.4), the color of the tape shall be natural.

3.6 Finish. A finish option shall be specified, in accordance with 3.6.1 or 3.6.2. Tape finishes shall not contain any mercury or copper. All finishes shall be able to be used freely in direct contact with insulated cable or wire.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&inspecomments@dla.mil . Since contact information can change, you may want to verify the currency of this address information using ASSIST Online database at http://assist.daps.dla.mil .

FSC 4020

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3.6.1 Finish A - Natural. No finish shall be added.

3.6.2 Finish C - Synthetic rubber. Tape shall be uniformly impregnated with a synthetic rubber or elastomer finish. Treated tape shall contain 7-17% rubber by weight. (see 5.2.1)

3.7 Deleted.

3.8 Put-up. Unless otherwise specified, the tape shall be furnished on parallel wind spools (reels) or universal wind tubes (holders). Size 2 tape shall be put-up in a minimum of 250 yard lengths, and size 4 and 5 tapes shall be put-up in a minimum of 500 yard lengths. The tape shall be free from twists, lumps, or projecting ends and shall be evenly wound so that each turn and layer is free from entanglement and twisting. There shall be no more than four pieces per reel or holder, and no piece shall be less than 50 yards in length.

3.9 Breaking strength Minimum breaking strength shall be as stated in Table 1 when tested as specified in FED-STD-191 Method 4108

3.10 Fungus resistance. No tape shall show visible growth (to the naked eye) of fungus on the surface of the test specimens when tested as specified in FED-STD-191 Method 5760.

3.11 Blocking. There shall be no visible damage or removal of the coatings on finish C tapes when tested as specified in 5.2.3.

3.12 Accelerated aging. When specified (see 7.4), finish C tapes shall show no evidence of stiffness, brittleness, softness, or tackiness when tested as specified in FED-STD-191 Method 5852, except that the specimen shall be six inches in length by the full width of the tape.

3.13 Identification. Tape shall be tagged with a label or ticket containing at least the following information: CID part number, National Stock Number, date of manufacture, and manufacturer's name. When put-up is non-standard, the tag shall indicate the actual length.

4. REGULATORY REQUIREMENTS

4.1 The offerer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Type 3 tape shall be accompanied by a warning label or sheet calling attention to the generation of hazardous vapors at temperatures above 400 degrees fahrenheit (204 degrees celsius).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producers own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

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5.2 Inspection requirements

5.2.1 Determination of percentage of finish. The percentage of finish C impregnating materials on the respective type tapes shall be determined during processing, by the manufacturer, by weighing an identical sample of tape or yarn (see 3.6), before and after the impregnating process. The test specimen shall be a minimum of 30 yards of tape or 120 yards of yarn as applicable. Weight (Length per pound) shall be determined in accordance with Method 4010 of FED- STD-191. The percentage of finish shall be calculated as follows:

$$\text{Percent Finish} \quad \frac{A-B}{A} \times 100$$

Where: A = Length per pound of untreated tape or yarn, as applicable.
B = Length per pound of treated tape or yarn, as applicable.

5.2.2 Deleted.

5.2.3 Determination of blocking. Ten turns of the tape shall be wound on a one-quarter inch diameter clean metal mandrel under a two pound tension and the end secured through holes in the mandrel. Eight turns of tape shall then be wound on top of the first layer under the two pound tension and the ends secured as before. The wound mandrel shall then be placed in a temperature controlled oven at 70 ± 1° C for two hours, and in such a manner that no part of the specimen comes in contact with the surface of the oven. The specimen shall then be removed and cooled at room temperature. After cooling, the outer layer shall be unwound and examined for evidence of damage to the coatings due to adhesion between layers of turns. The first layer shall be examined while still in place for similar evidence of adherence and damage.

6. PACKAGING.

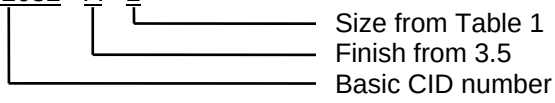
6.1 Preservation, packing and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Intended use. The tape is intended for lacing and tying electrical wire and cable bundles. Tape with finish C provides the best knot holding characteristics. TFE-fluorocarbon may provide good resistance to fluids and fuels. Useful temperature range of TFE-fluorocarbon tape is -100 °F to 450 °F.

7.2 Part Identification Number (PIN). The PIN shall consist of the basic CID number, followed by the finish code letter from 3.6, followed by the size number from Table 1.

PIN Example: AA52082 - A - 2



7.3 Source of documents.

7.3.1 Government documents. FED-STD-191 “Textile Test Methods” is available from the Navy Publications and Printing Service Office, Standardization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or at http://assist.daps.dla.mil.

7.4 Ordering data.

a. Title, Number and Date of this Commercial Item Description.
b. Size and finish required (see 3.3 and 3.6).
c. Color, if other than specified (see 3.5).
d. Put-up if other than specified (see 3.8).
e. When accelerated aging is required (see 3.12).

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7.5 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. This list may not be indicative of all possible NSNs associated with the CID.

NSN	Type	Size	Finish
4020-00-809-9407	3	4	A
4020-01-446-4647	3	4	C

7.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - AS, MC, OS
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CIVIL AGENCY COORDINATING ACTIVITIES:
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